

Big Data Systems Final Project: AI Prompt Monetization System

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GitHub Repository : https://github.com/Vaporeon-04/BigData2026_Final_Project/tree/main

Live Deployment Demo : <https://bigdata2026finalproject-ddtiutjmppfbvzwhjhdq2d.streamlit.app/>

1. Target Customer

Our system serves a dual-track market (B2C and B2B), focusing specifically on AI image creators and commercial art teams who struggle with anatomical inconsistencies in AI generation.

1.1 Target 1: Independent Creators & Commission Artists (B2C)

- **Target Persona:** Individual artists operating on social media, Patreon, or handling private commissions who utilize AI to boost their production speed.
- **Current Workaround:** When encountering AI generation flaws (e.g., mutated hands, asymmetrical eyes, twisted limbs), they currently waste hours manually "inpainting" or blindly re-rolling prompts through trial and error.
- **Why Our System is Better:** We offer a micro-subscription model that provides instant access to a verified "Anatomy-Fixing Prompt Database." Instead of guessing, they can query our database to retrieve highly detailed, data-backed long-tail prompts and negative prompts that are mathematically proven to generate stable anatomy, saving them hours of manual labor.

1.2 Target 2: Indie Game Studios & VTuber Agencies (B2B)

- **Target Persona:** Small to medium commercial teams requiring a massive volume of highly consistent visual assets (character sprites, CGs) but lacking the budget to hire dedicated AI prompt engineers.
- **Current Workaround:** Relying on expensive traditional outsourcing or struggling with inconsistent AI art that breaks immersion across different game scenes.
- **Why Our System is Better:** We offer an Enterprise License and an Automated Generation API. They are not just buying prompts; they are buying an integrated pipeline. By connecting to our API, they can automatically inject our top-tier parameters into their workflow, ensuring commercial-grade consistency without needing in-house prompt expertise.

2. Evidence of Demand and Willingness to Pay

2.1 Data Acquisition Process & Evidence of Demand

To prove that anatomical inconsistency is a real, monetizable pain point, we developed a web scraper targeting major AI art discussion forums (e.g., PTT [AI_Art](#) boards).

- **Methodology:** We scraped historical posts and analyzed keyword frequencies to identify the most common generation failures, as shown in Fig. 1.

- **Findings:** The data highlighted specific anatomical keywords. "Hands", "Eyes", and "Anatomy collapse" were the most frequently discussed bottlenecks. The data proves that the market actively seeks solutions for these specific generation flaws.

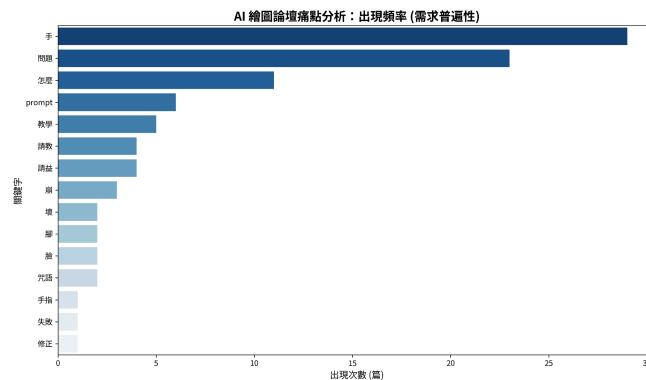


Fig. 1 frequencies of keywords that identify common generation failures on PTT

2.2 Willingness to Pay & Pricing Benchmarks

We quantify the customer's willingness to pay through both time saved and competitive benchmarking:

- **Non-Monetary Value (Time Saved):** Resolving a "bad hands" generation typically costs an artist 1 to 2 hours of manual fixing. Our system reduces this to a 5-second search query.
- **Monetary Pricing Strategy:**
 - **Freemium:** 3 free searches/day to build user habits.
 - **Pro Subscription (B2C):** \$150 NTD/month for unlimited queries. Priced similarly to basic creative tools (like Canva Pro), making it an easy impulse buy for independent artists.
 - **API Usage-Based (B2B):** \$300 NTD per 10,000 API calls for indie game studios needing automated asset generation.
 - **Enterprise License (B2B):** \$50,000+ NTD/year for unlimited database access and guaranteed SLA, targeting well-funded commercial teams.

3. System Design Expectations

3.1 Data Sources

Our system ingests two distinct types of data to bridge market demand and product supply:

- **Demand Validation Data:** We built web scrapers to ingest historical forum discussions from PTT. This unstructured text data helps us identify trending anatomical pain points.
- **Product Supply Data:** To overcome API rate limits, we ingest the [poloclub/diffusiondb](#) dataset directly from Hugging Face via Parquet format, providing high-quality AI prompts and their corresponding image URLs.

3.2 Storage and Processing

While our local prototype utilizes Pandas and local CSVs for rapid iteration, our production architecture is designed around distributed Big Data paradigms:

- **Storage:** Raw HTML files and massive Parquet files are stored in a distributed object storage system (e.g., Amazon S3 or HDFS). The refined, searchable prompt data is loaded into **Elasticsearch** (NoSQL), which is superior for lightning-fast, fuzzy full-text searches.
- **Processing:** We utilize **Apache Spark** for our batch processing pipeline. The Spark jobs perform data cleaning and quality filtering based on our heuristics (**length > 100**, **steps >= 30**), ultimately outputting the "Premium" tier dataset.

3.3 Delivery

Customers consume our refined data through a multi-channel delivery system:

- **Interactive Web Dashboard (B2C):** Built using Streamlit, featuring dynamic image rendering and an interactive "Action Link".
- **Live Inference Integration:** The dashboard integrates with the [Pollinations.ai](#) HTTP GET API. When a user clicks "Generate", the system URL-encodes our verified prompt and triggers a live inference.
- **REST API (B2B):** For indie game studios, we expose a structured REST API that returns our verified JSON prompts.

3.4 Architecture Diagram

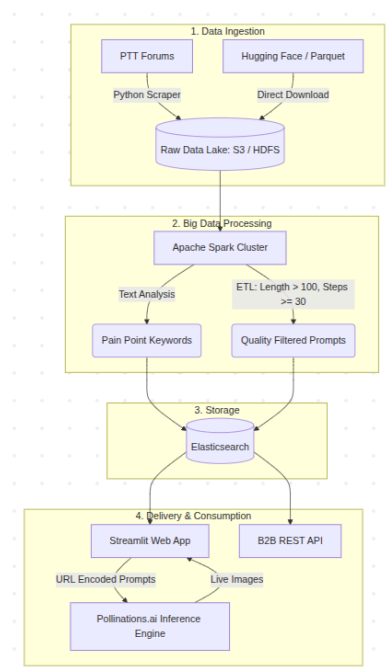


Fig. 2 The flow chart of our architecture

3.5 Scalability & Cost (At 100x Scale)

If our user base and data volume grow by 100x:

- **Compute:** We would migrate our processing to a managed Spark cluster (e.g., AWS EMR). Cost scales linearly, but batch processing can utilize cheap Spot Instances.
- **Delivery:** We would containerize the application using Docker and deploy it on a Kubernetes cluster with a load balancer.
- **Caching:** To minimize expensive database queries and third-party API calls, we would introduce a **Redis** in-memory cache to instantly serve the most frequently searched keywords.

4. Go-to-Market Difficulties

4.1 Cold-Start Problem & Data Acquisition Costs

The primary hurdle for this system was the "Cold-Start Problem": to sell high-quality prompts, we first needed to acquire them. Our initial approach was to scrape the Civitai API. However, we encountered strict platform restrictions, missing API keys, and creators intentionally hiding their metadata to protect their unique styles. As a consequence, instead of engaging in an arms race with Civitai's anti-scraping mechanics, we pivoted to open-source, commercially viable datasets. We utilized the Hugging Face [poloclub/diffusiondb](#) Parquet dataset, directly ingesting raw metadata to bypass script-blocking security policies.

4.2 Trust and Adoption (Quality Assurance)

Why would a customer trust our paid database over a free search? To build a competitive moat, we implemented a rigorous Automated Quality Validation Pipeline: we do not sell raw, unverified data. Our processing layer filters the massive dataset using strict heuristics—retaining only generation parameters with high iteration steps (**step >= 30**) and substantial detail length (**prompt length > 100 characters**). This ensures that every prompt in our marketplace is a "verified long-tail prompt" backed by data, overcoming the trust barrier and justifying our subscription fee.

5. Data Limitations and Future Enhancements

In a real-world commercial pipeline, pairing a prompt with its specific Checkpoint model and LoRA (Low-Rank Adaptation) network is crucial for guaranteeing stylistic consistency. However, a key limitation we encountered during the data ingestion phase is that our primary verified data source, the [poloclub/diffusiondb](#) dataset, consists of early-generation Stable Diffusion data. It inherently lacks native metadata fields for specific Models and LoRAs.

Rather than synthetically fabricating these fields for the prototype, we chose to maintain strict data integrity. We acknowledge this as a real-world data acquisition obstacle. To solve this in future production iterations, the system's ingestion layer will be upgraded to negotiate commercial API access with modern model repositories (such as Civitai's enterprise tier) to secure complete, end-to-end generation metadata.